

MODULE 22

SQL Security and User Management

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

SQL Security and User Management

User Management

```
CREATE USER 'arshith'@'localhost' IDENTIFIED BY 'Pass123!';  
ALTER USER 'arshith'@'localhost' IDENTIFIED BY 'New!';  
DROP USER 'arshith'@'localhost';  
SELECT user, host FROM mysql.user;
```

GRANT Privileges

```
GRANT ALL PRIVILEGES ON school.* TO 'arshith'@'localhost';  
GRANT SELECT, INSERT ON school.students TO 'teacher'@'localhost';  
GRANT SELECT ON school.* TO 'reporter'@'localhost';  
FLUSH PRIVILEGES;
```

REVOKE Privileges

```
REVOKE DELETE ON school.students FROM 'teacher'@'localhost';  
REVOKE ALL PRIVILEGES ON school.* FROM 'teacher'@'localhost';  
SHOW GRANTS FOR 'arshith'@'localhost';
```

SQL Injection

Vulnerability: WHERE user = "" + input + ""

Attack input: admin' OR '1'='1 --> bypasses login!

Prevention:

Use prepared statements / parameterized queries

Validate and sanitize all user input

Least-privilege database accounts

Escape special characters

Security Best Practices

Never store plain-text passwords (use bcrypt)

Encrypt sensitive columns at rest

Regular automated backups

Audit logging on sensitive tables

Use VIEWS to limit data exposure

Principle of least privilege for every user